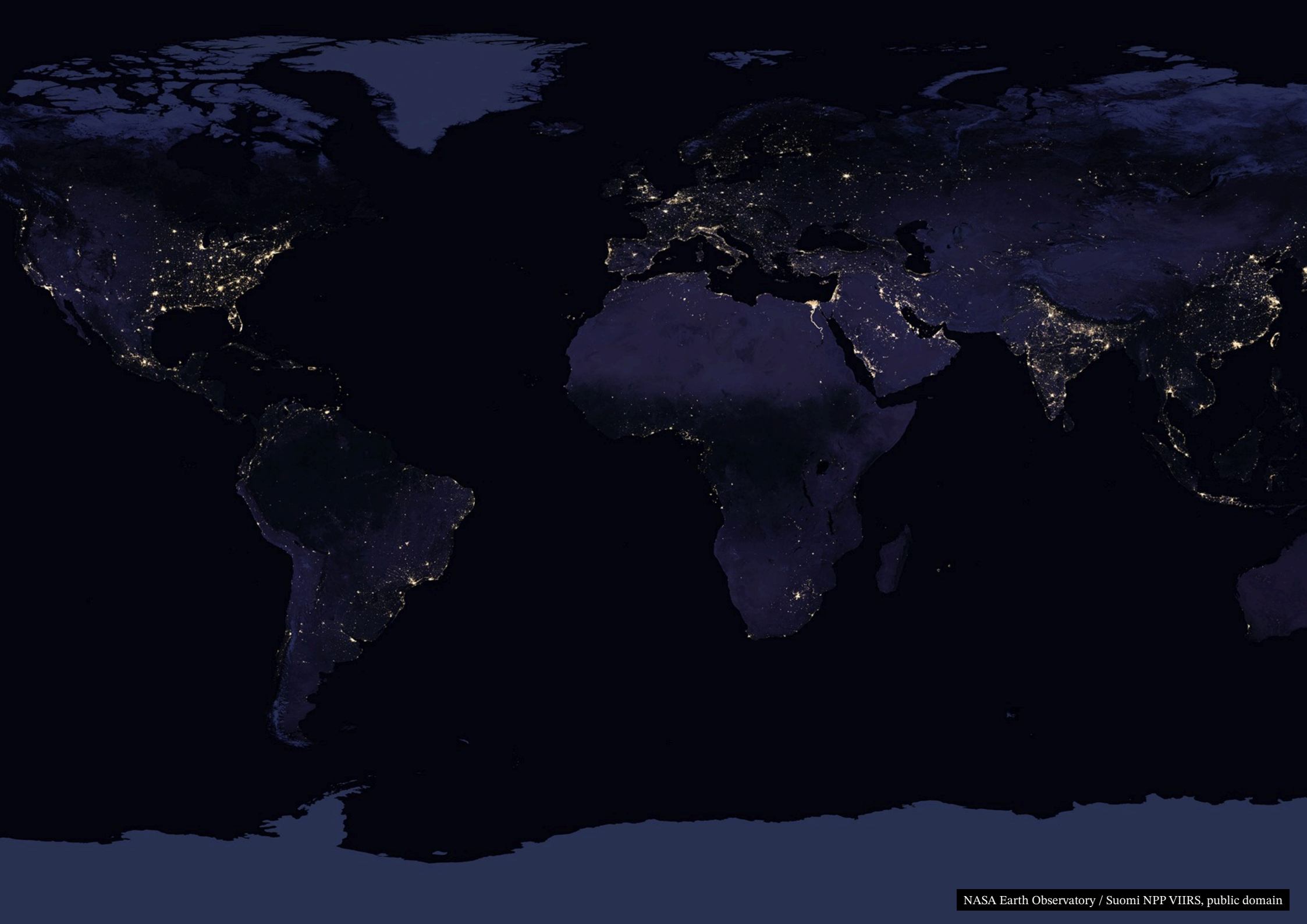
Fusion Physics · Chapter 0

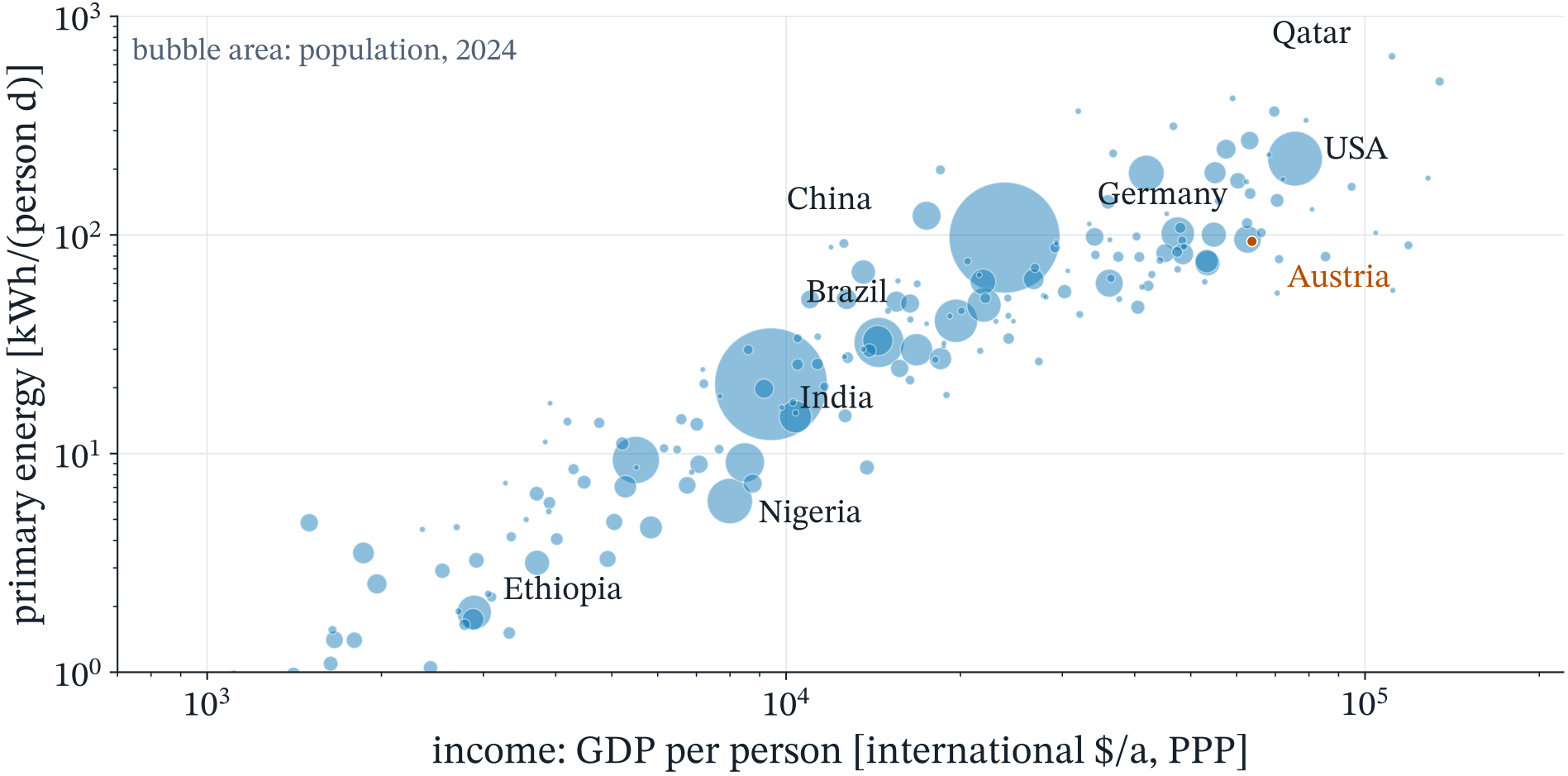
The big picture

Christopher Albert



NASA Earth Observatory / Suomi NPP VIIRS, public domain

0.1 Energy and income




Monthly income: PPP US dollars per equivalent adult · Dollar Street
Haiti 40 $

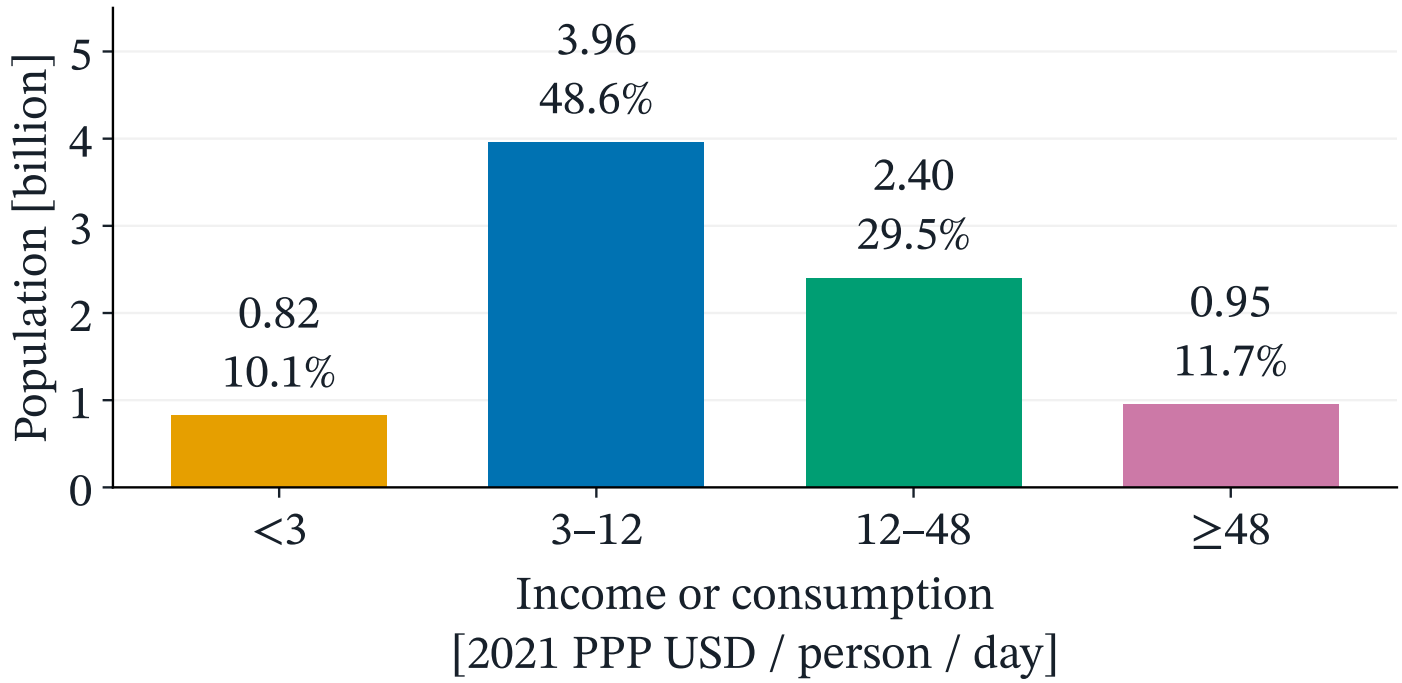

Dollar Street, Gapminder: Z. Miller (Haiti, Bolivia), L. Carvalho (Brazil), O. Havana (Spain); CC BY 4.0
Bolivia 180 $


Brazil 685 $


Balay
Spain 7639 $

0.1 Living standards · 2024

World population · 8.14 billion



91.9%

Electricity access
world

≈ 50 ×

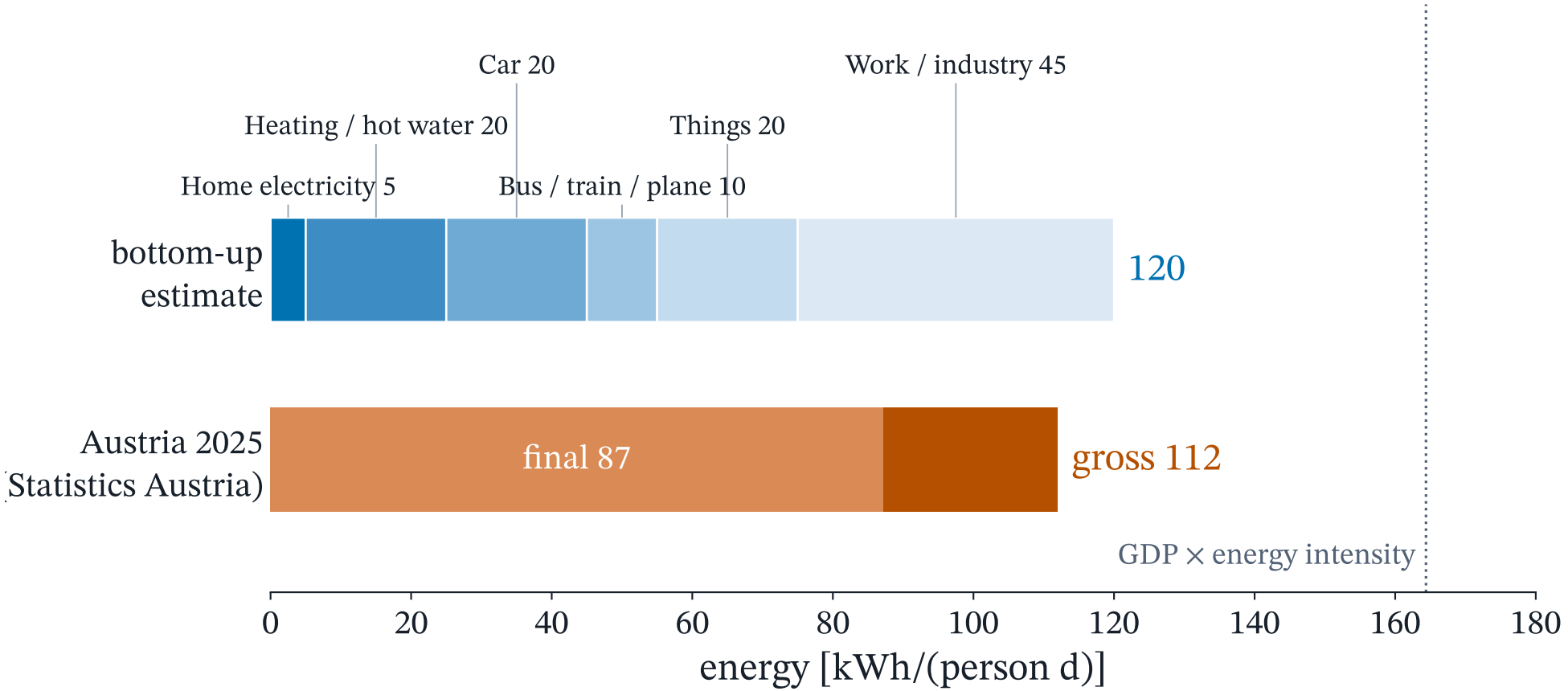
Primary energy
per person
Austria / Ethiopia

Source: World Bank PIP · Sep 2026 · global estimates

Electricity access → lighting, refrigeration, cooking, heating?

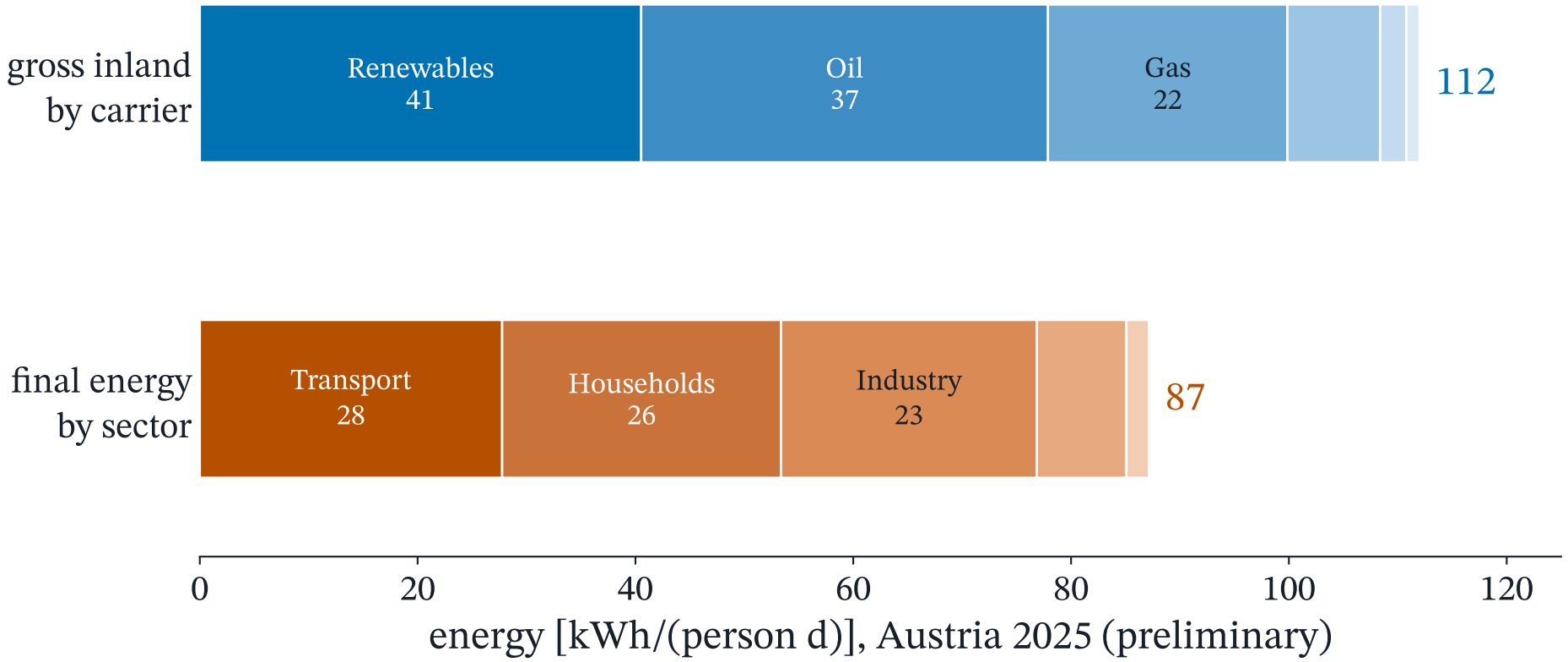
World Bank PIP, Sep 2026 · 2021 PPP · fixed bands: 3 / 12 / 48 USD/(person day)
WDI electricity access · EIA primary energy: traditional biomass incomplete

0.4 Micro and macro



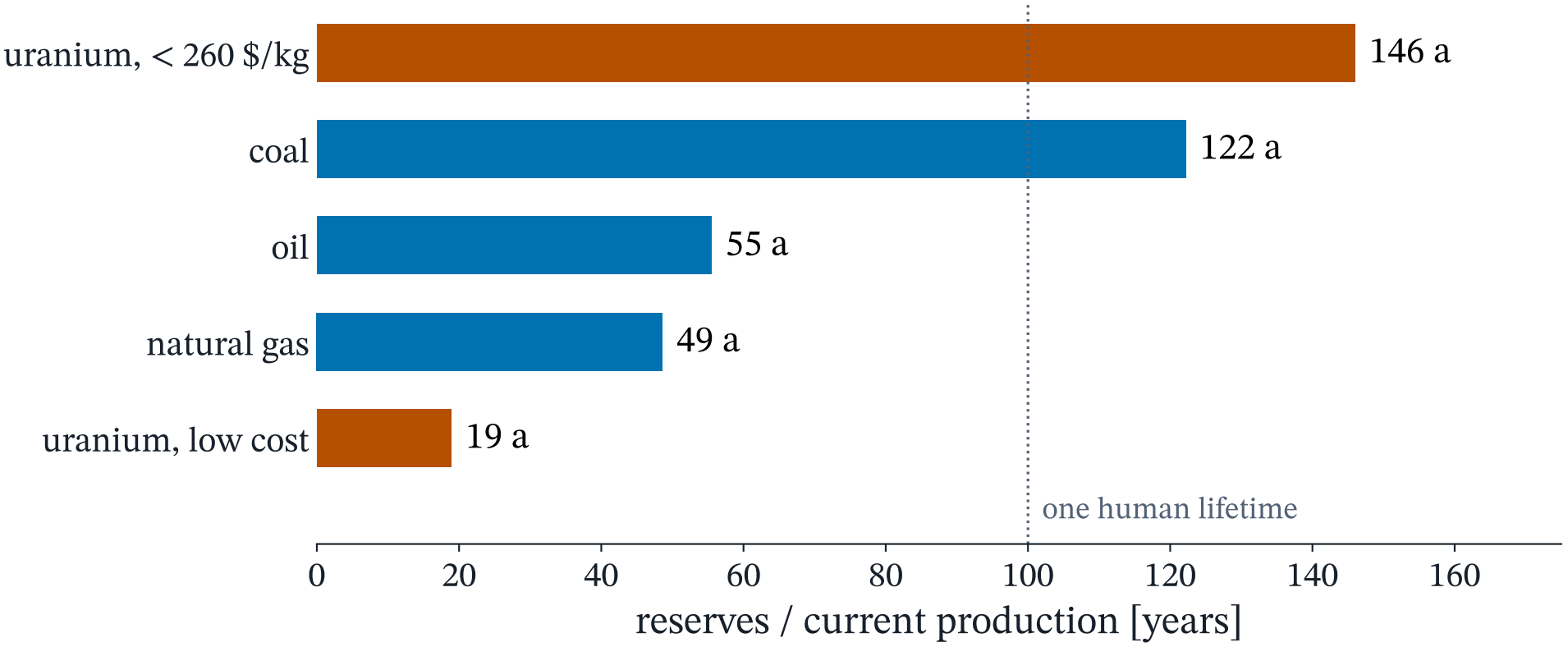
gross inland = primary; final = delivered to users; Austria 2025, Statistics Austria (preliminary)

0.4 Austria’s energy mix



gross inland − conversion and grid losses − non-energy use = final energy; Statistics Austria 2025

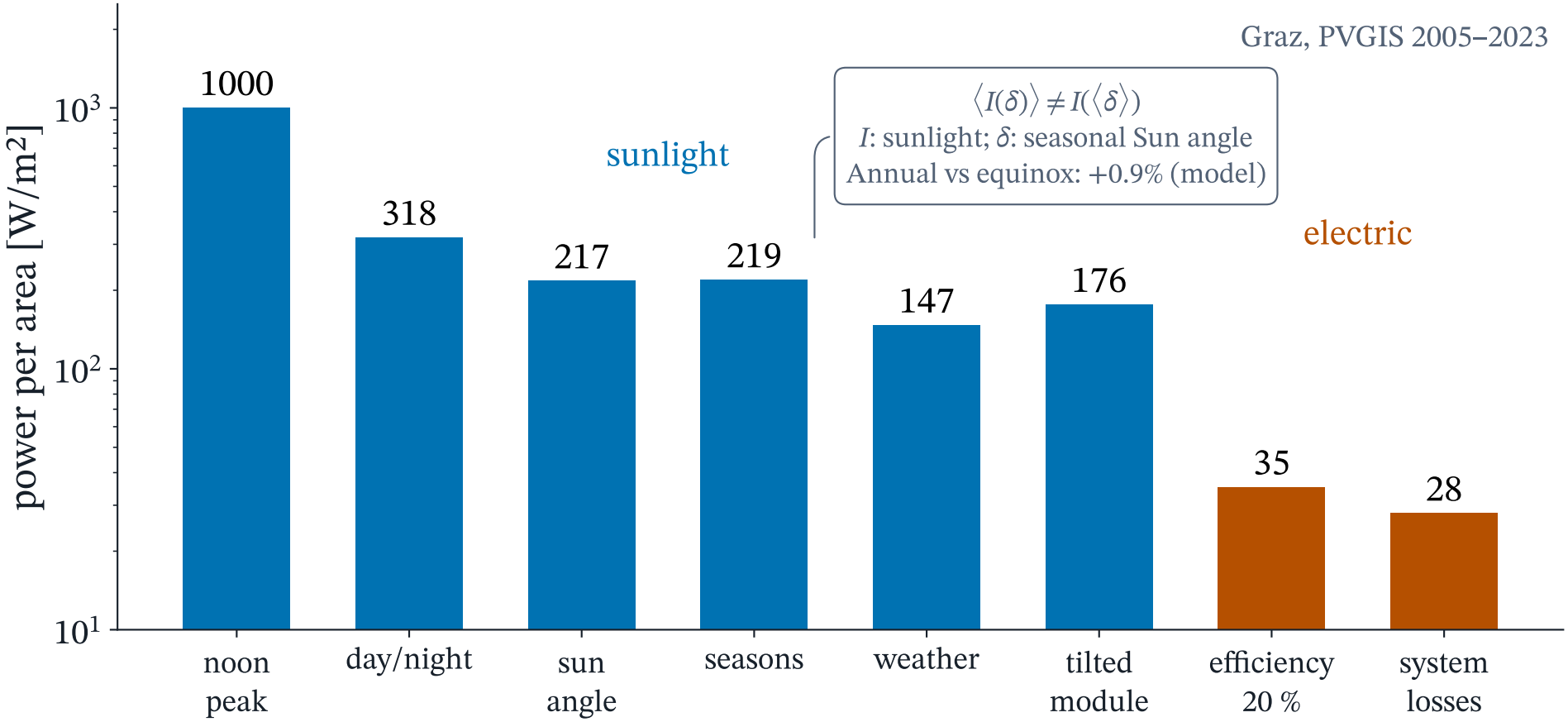
0.5 How long do reserves last?



Each fuel at its current world production rate · constant

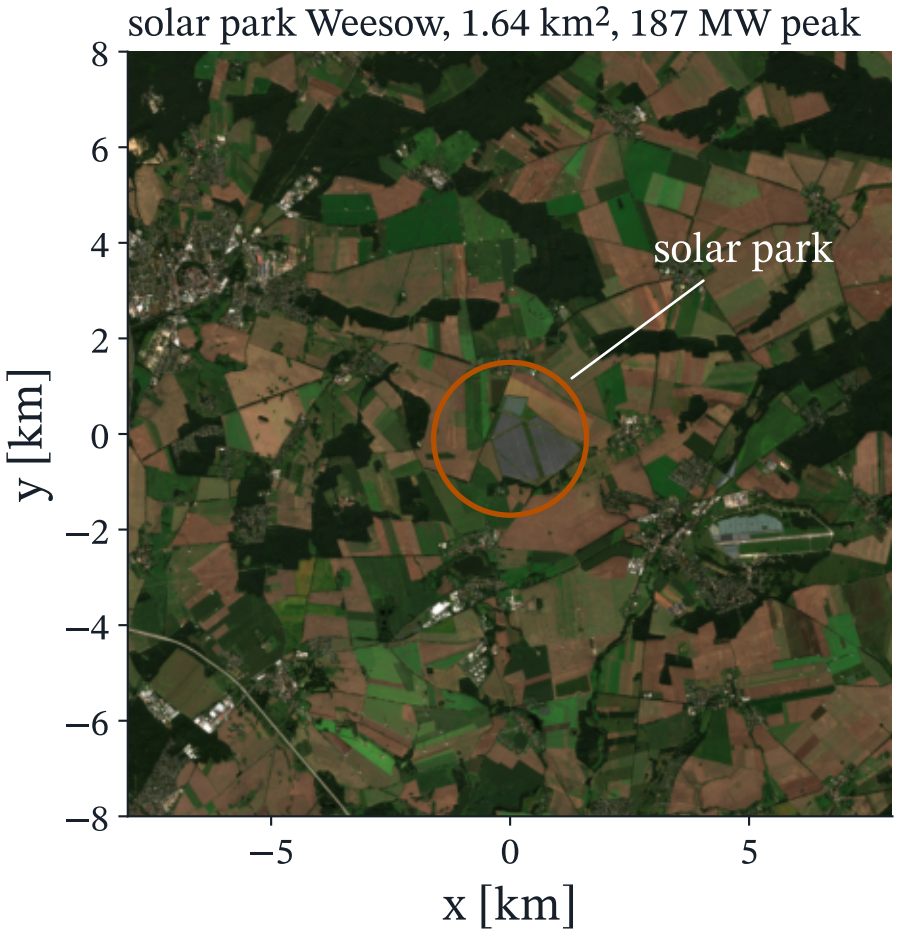
BGR: 2024 production; broad uranium resources: 2023 production, IAEA/NEA Red Book.
No demand growth or reserve additions; each fuel retains its own present share.

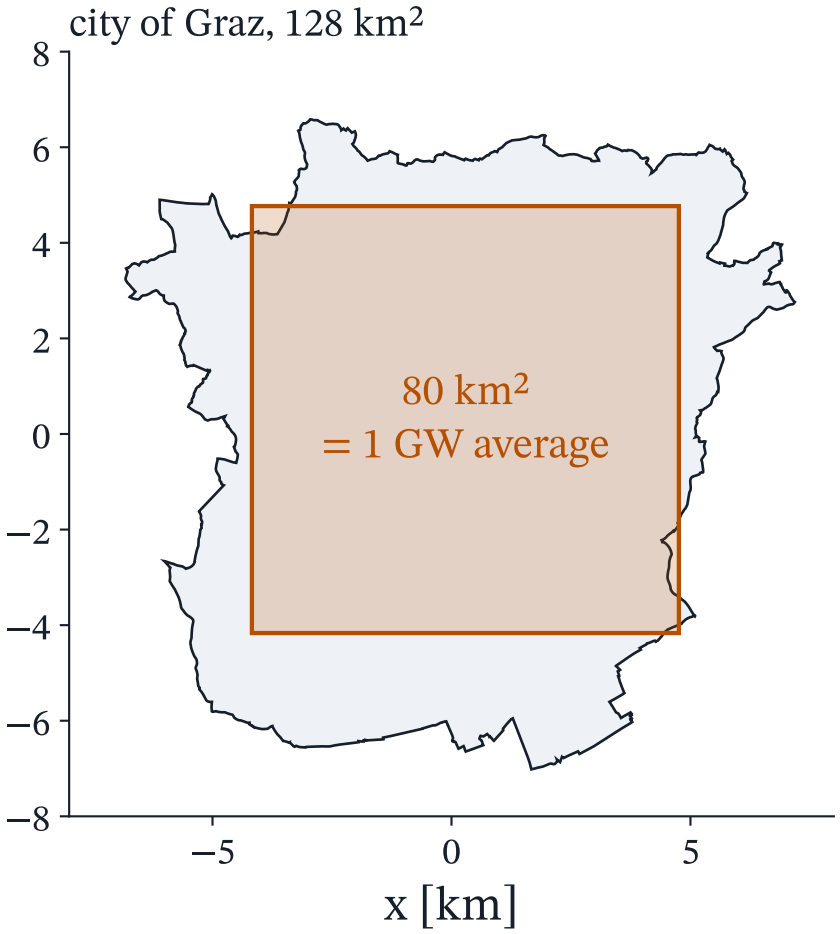
0.6 Solar power per area in Graz



PVGIS (EU JRC) for Graz; module at 20 % efficiency

0.6 How big is a gigawatt?



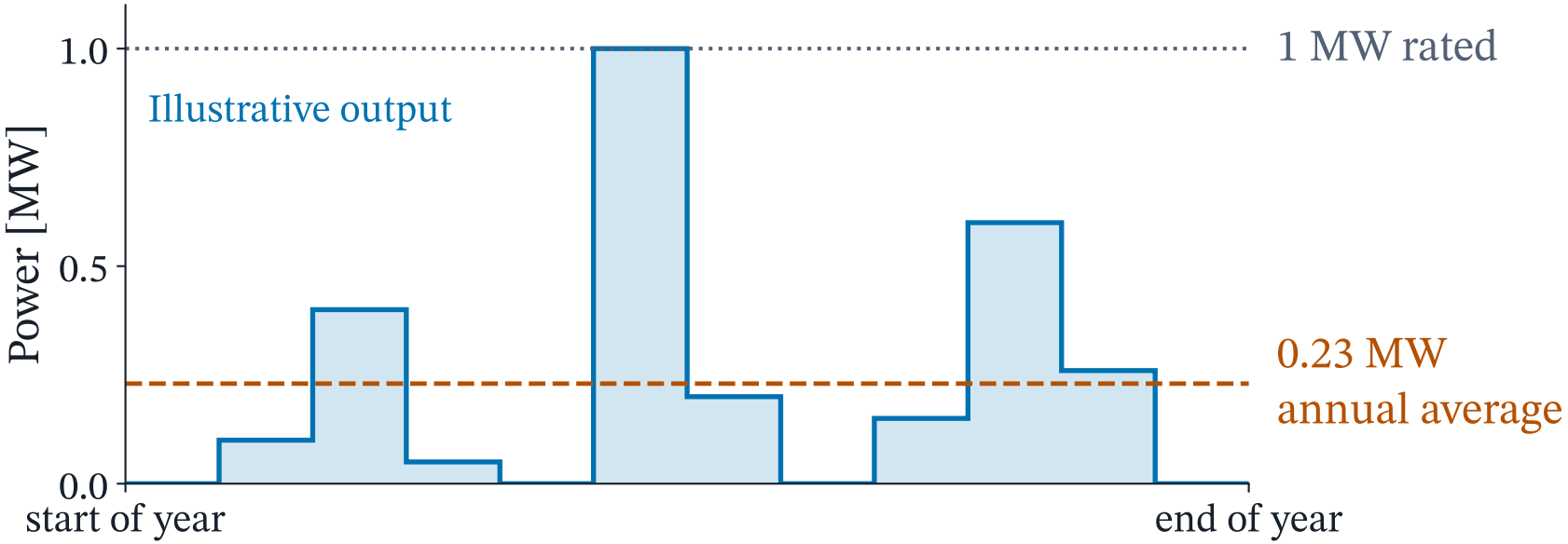


Contains modified Copernicus Sentinel data 2025 (Sentinel-2A, 11 Sep 2025); Graz boundary: Stadt Graz, data.graz.gv.at, CC BY 4.0; park data: EnBW (2021), expected yield 180 GWh/a.



Naturpuur, Wikimedia Commons, CC BY-SA 4.0

0.6 Wind power per area

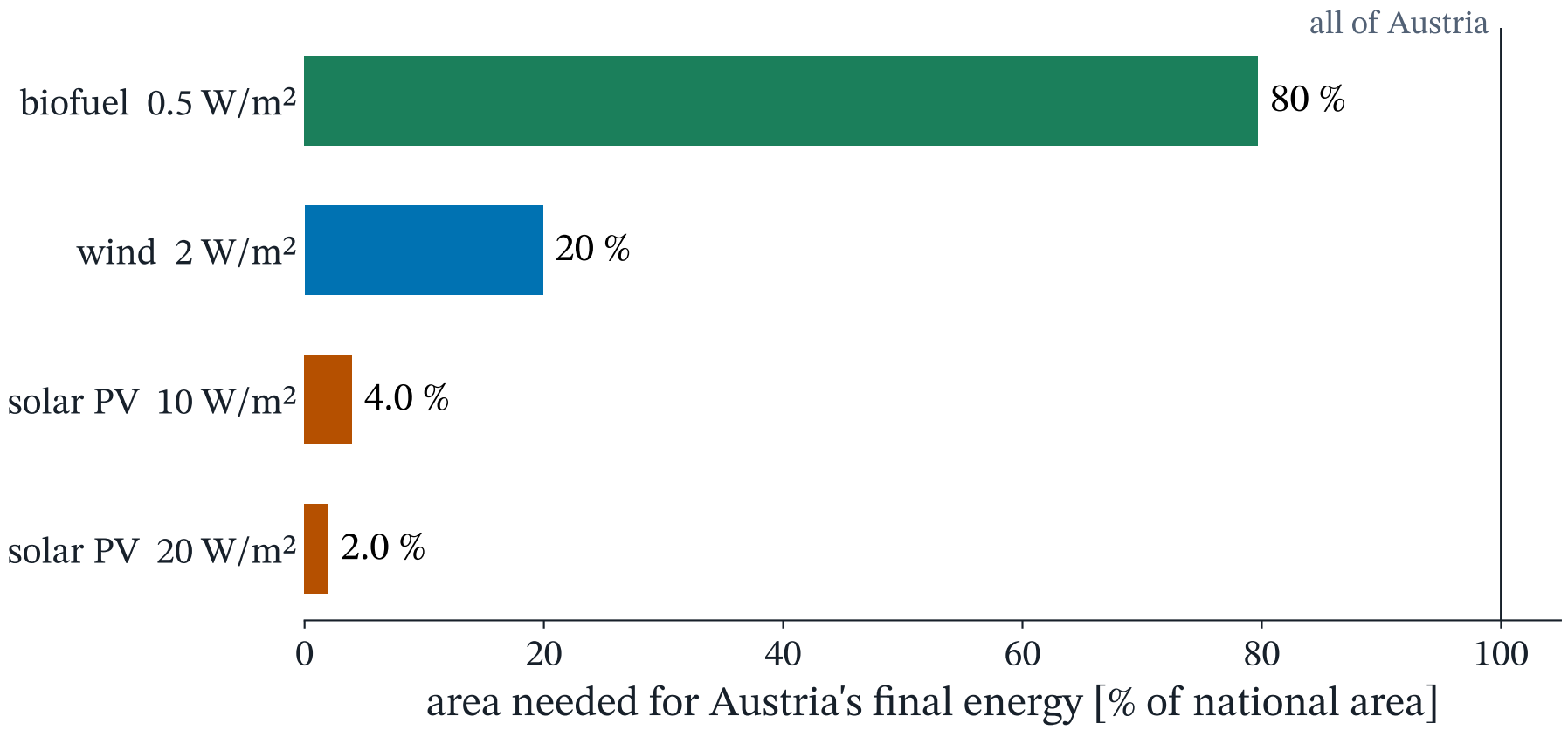


$$\text{capacity factor} = \frac{P_{\text{average}}}{P_{\text{rated}}} = \frac{0.23\ \text{MW}}{1\ \text{MW}} = 23\%$$

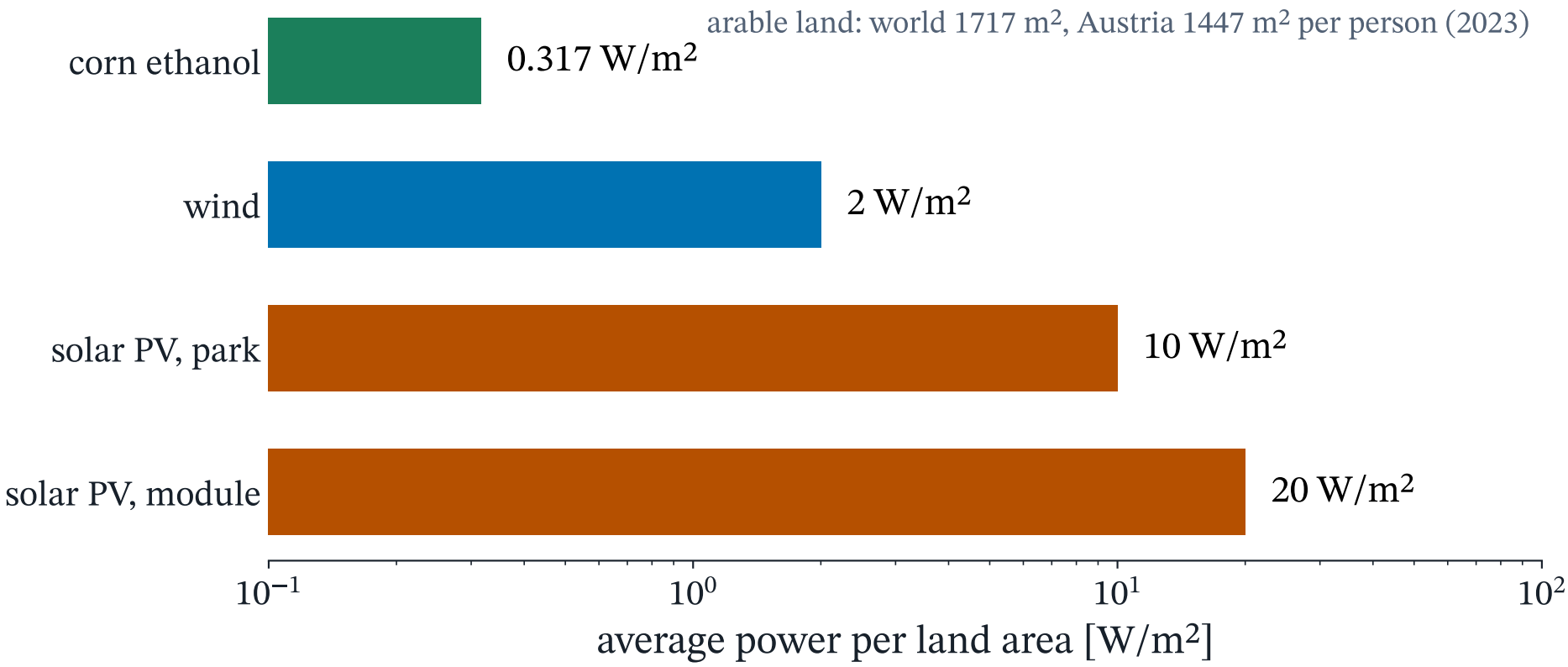
Annual-average power per wind-farm area

$$\frac{P_{\text{average}}}{A} \approx 1 \text{ to } 2 \frac{\text{W}}{\text{m}^2}$$

0.6 Land for Austria’s energy

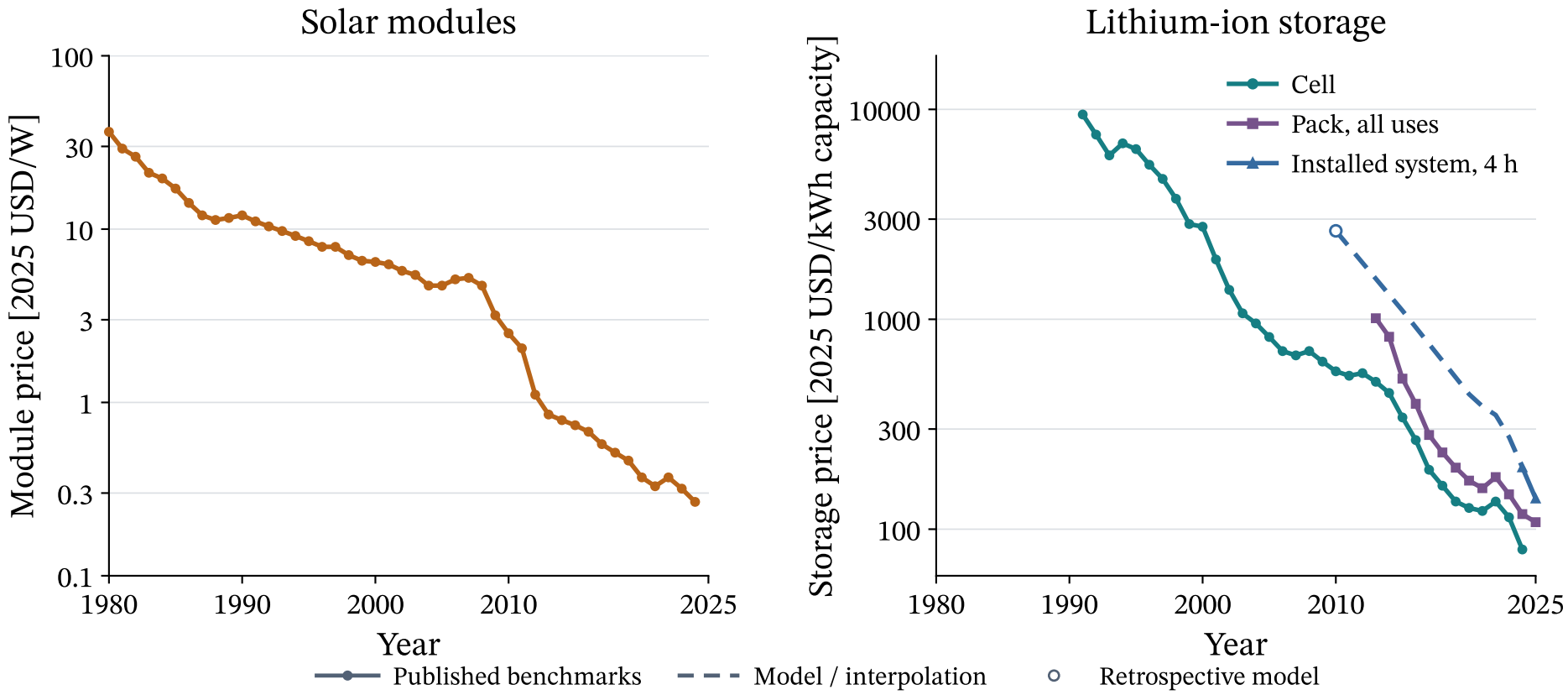


0.6 Food or fuel?



FAO arable land via World Bank; USDA corn yield, DOE ethanol energy; wind and solar: stated scenarios

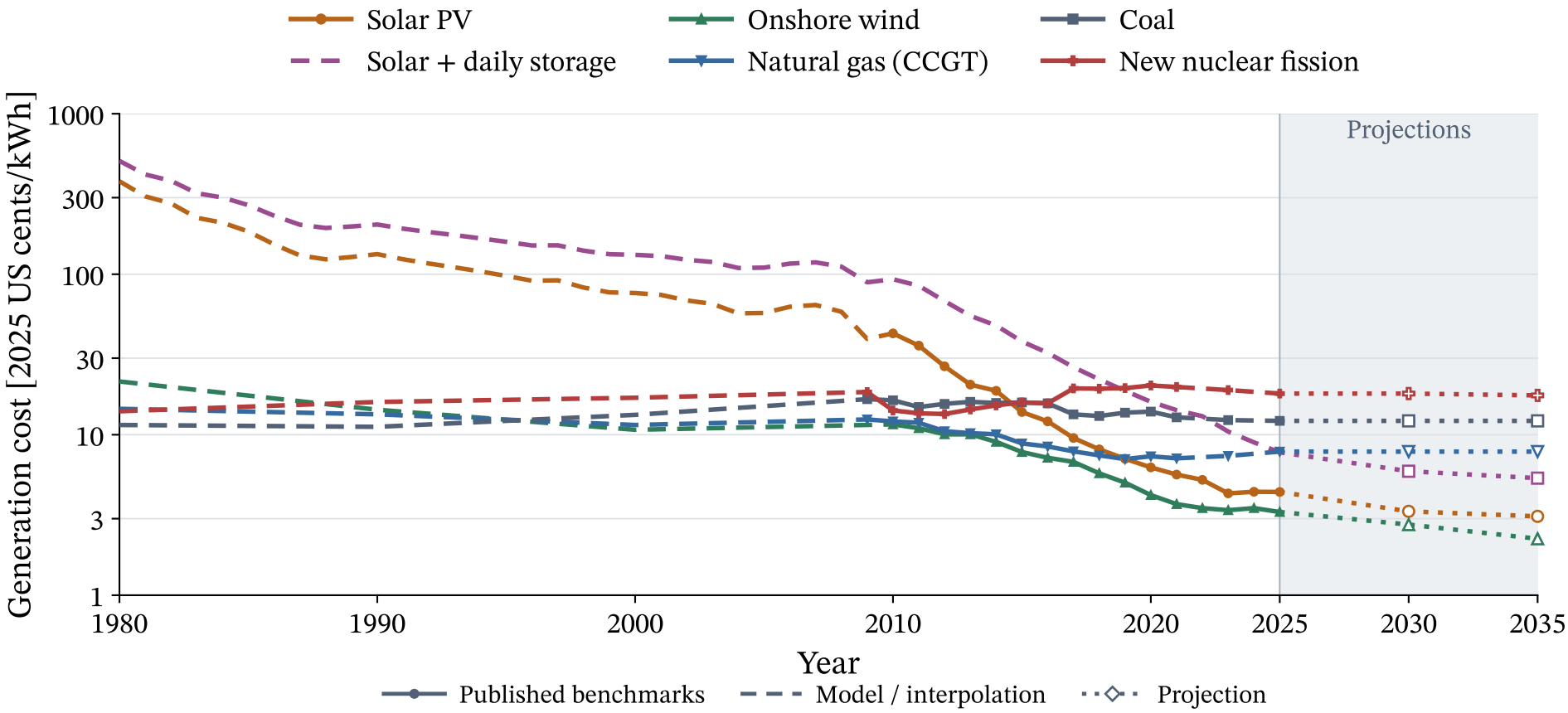
0.7 Solar panels and battery storage



Module and cell histories: OWID source series · Packs: BloombergNEF · Installed systems: IRENA

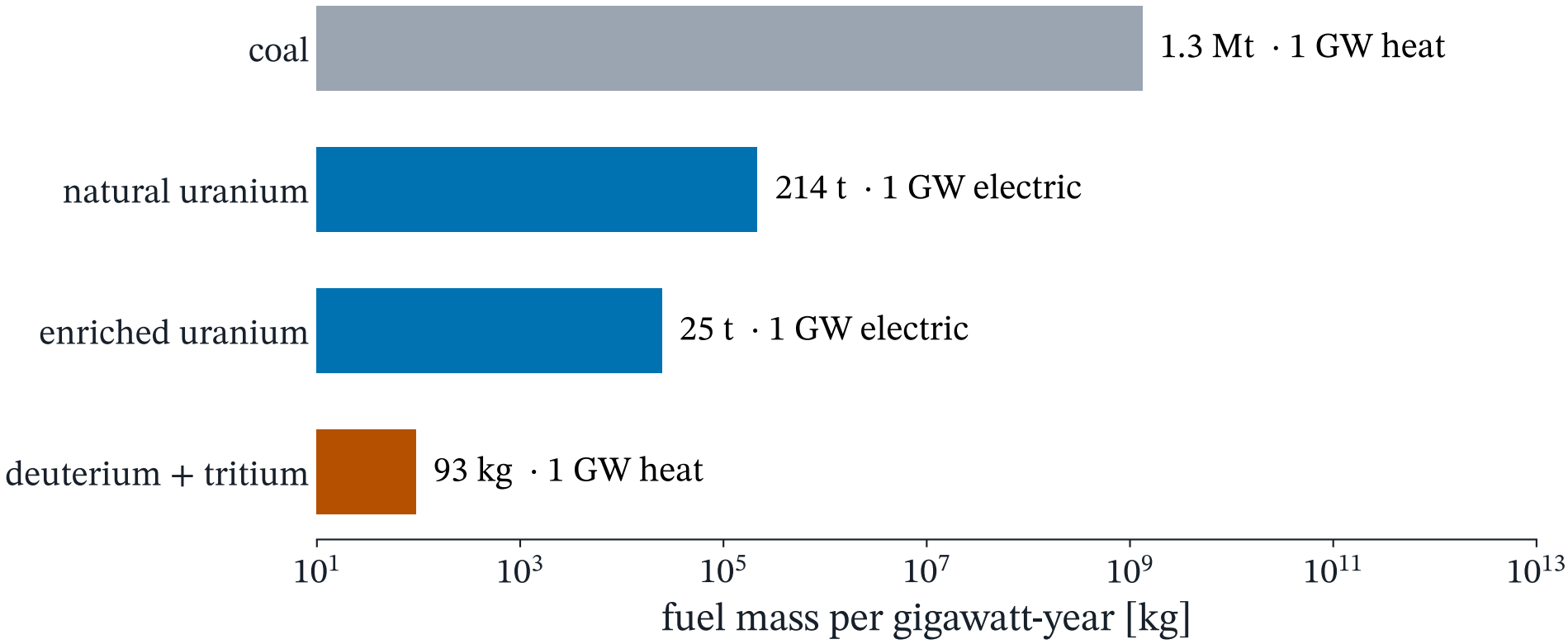
Sources: OWID module prices, cell prices, BNEF packs, IRENA installed storage.
Supplied sourcebook, 9 Oct 2026 · constant 2025 USD · cell, pack and installed-system cost boundaries

0.7 Electricity generation costs



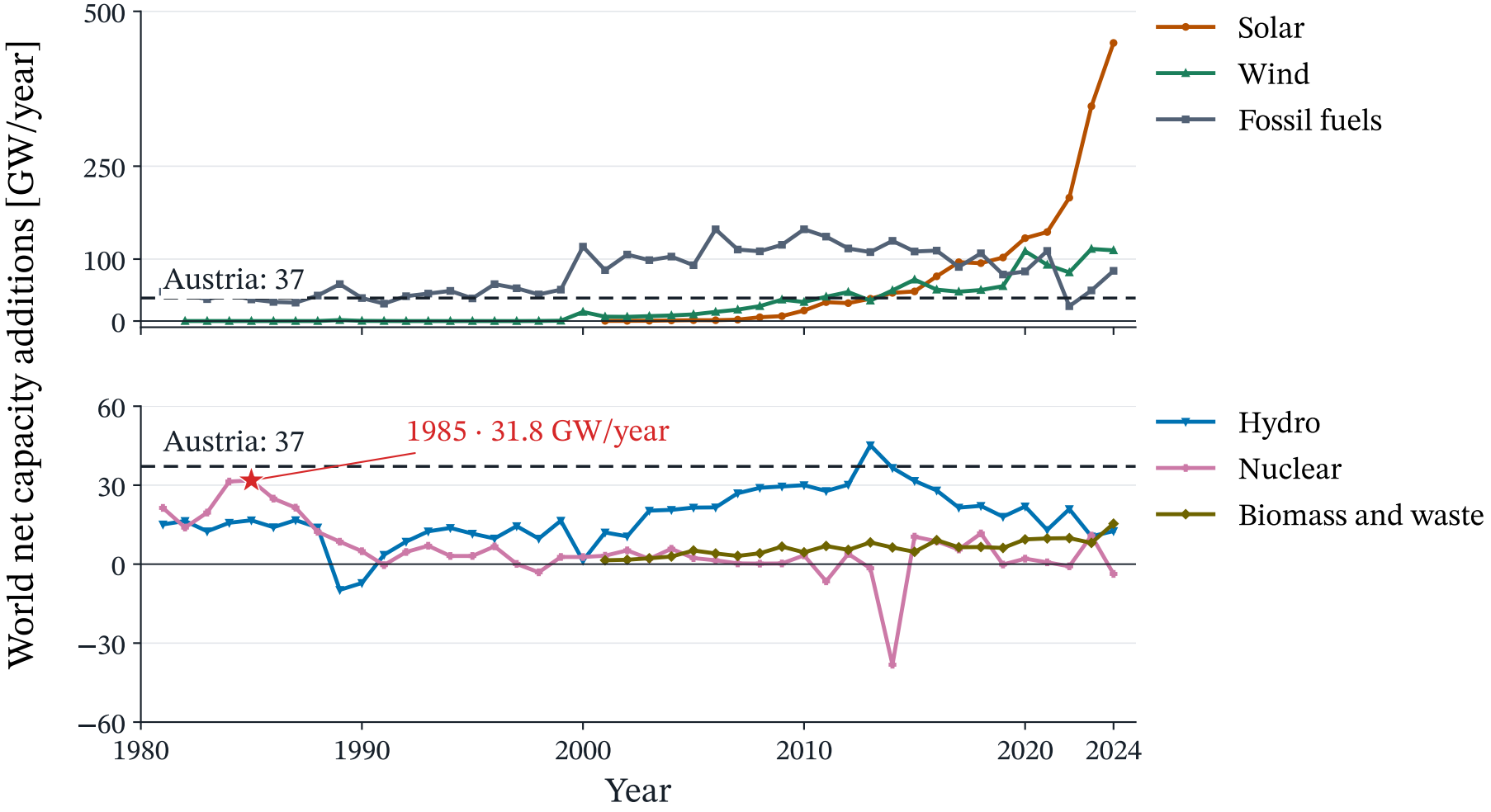
IRENA · Lazard · ATB trends, 2025 anchors (nuclear: flat to 2030)
Gas/coal: constant 2025 real cost · Daily storage: 60% shifted

0.8 Fuel for one gigawatt-year



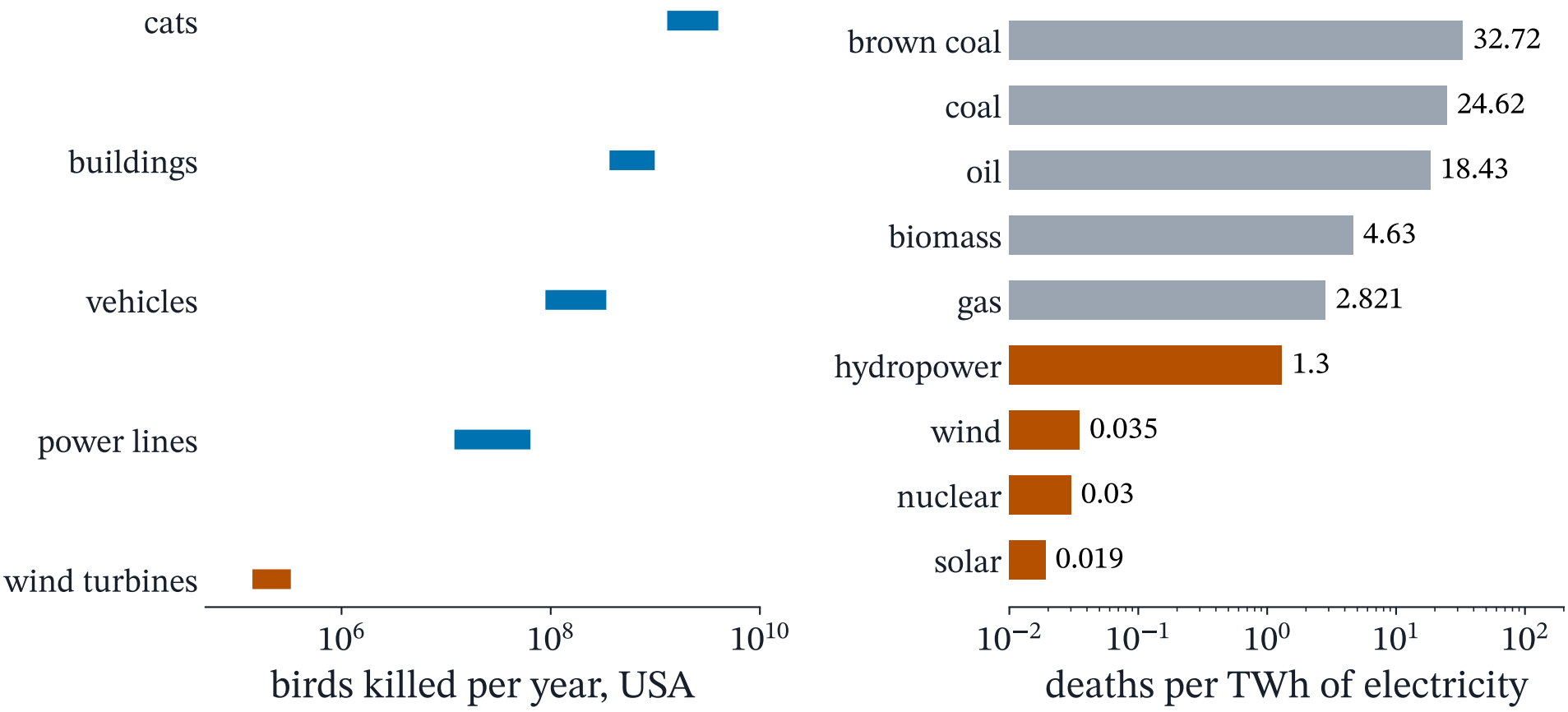
D + T → ^{4}He + n + 17.6 MeV; coal 24 MJ/kg; uranium once-through

0.8 Annual capacity additions



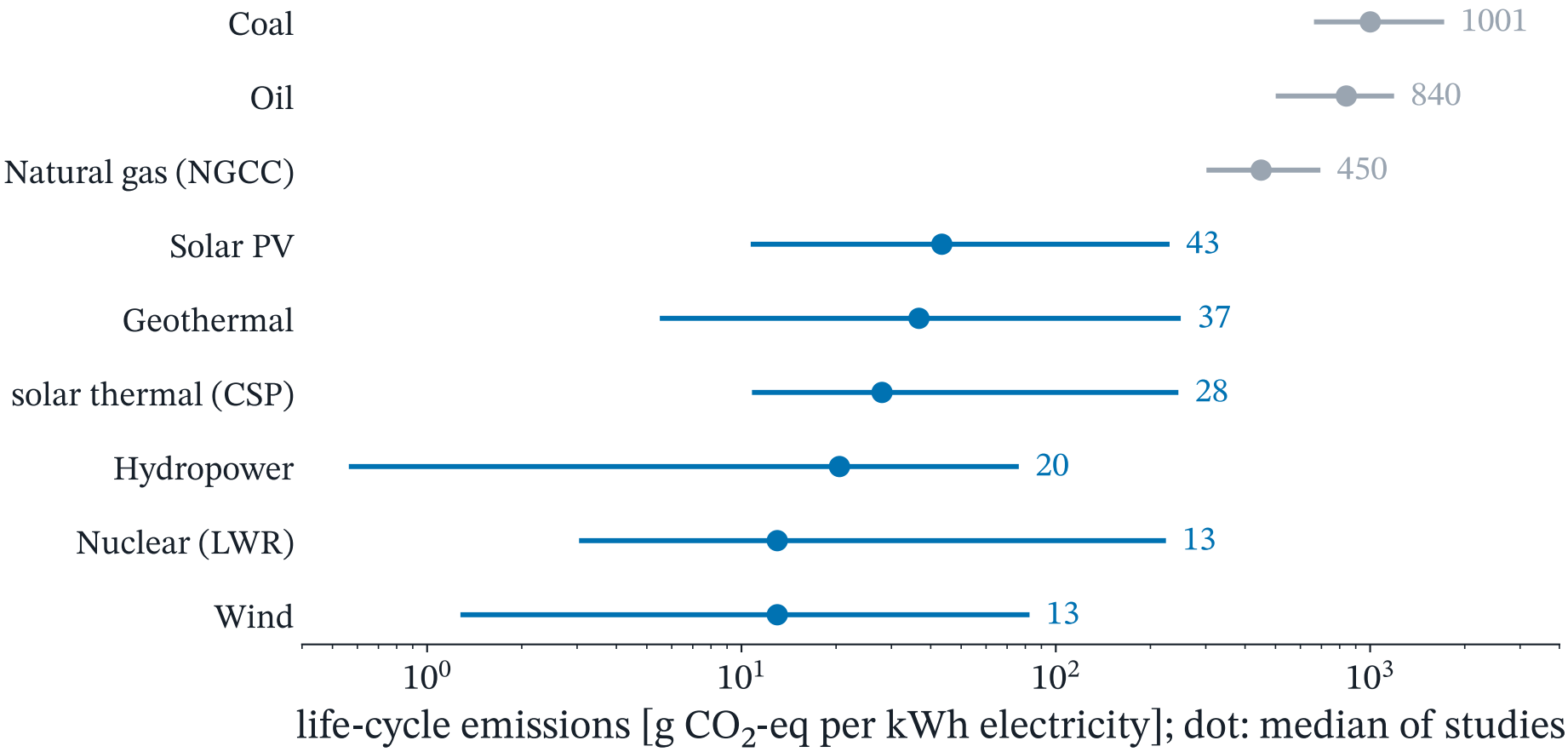
EIA · annual changes in installed capacity · Austria: final energy, 90% capacity factor, 1-year build

0.9 Perceived and measured risk

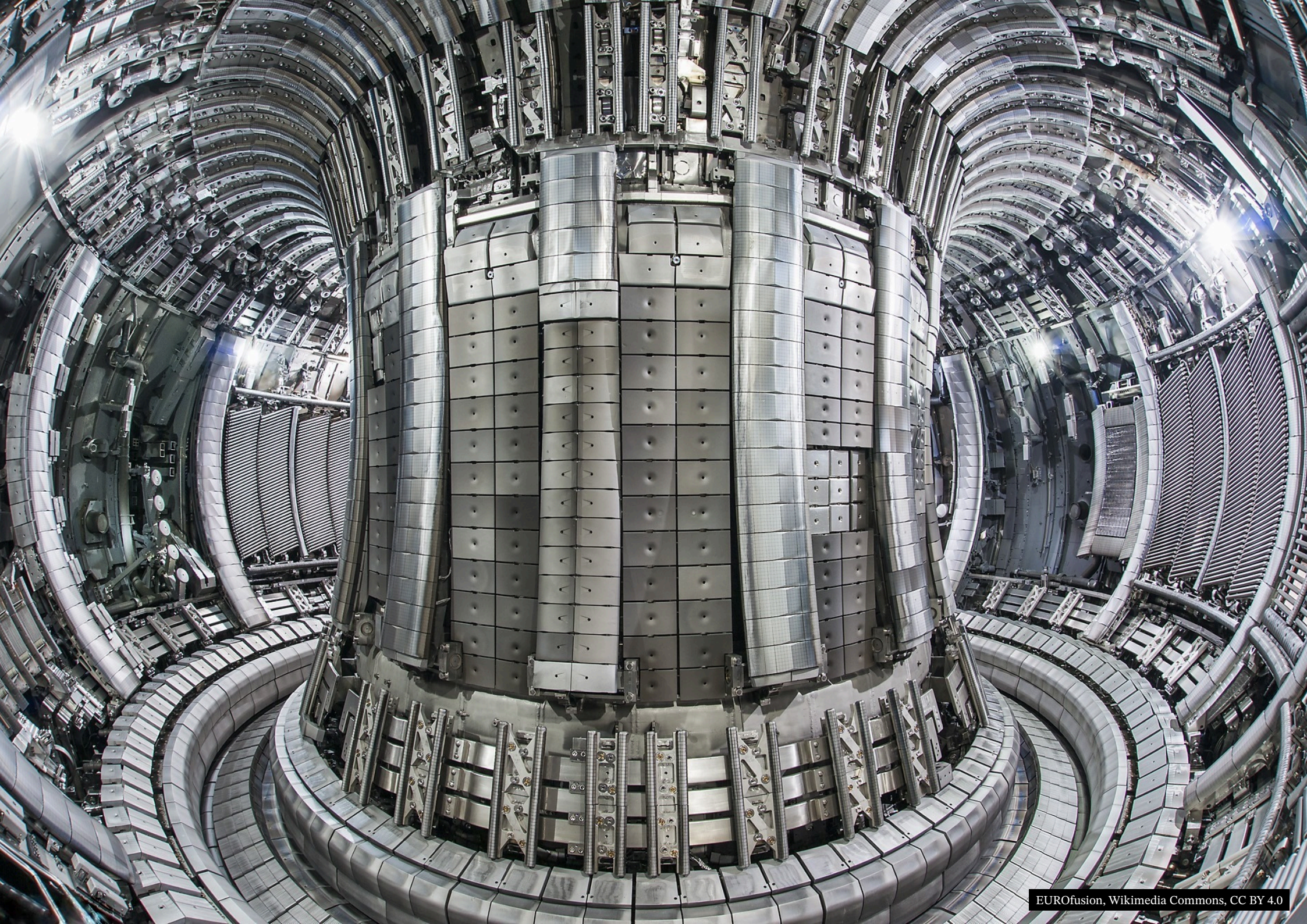


birds: Loss et al. (2013, 2014), USA; deaths: Markandya & Wilkinson (2007), Sovacool et al. (2016), via Our World in Data

0.10 Is anything CO_2-free?



DOE/NREL/ALLIANCE Life Cycle Emissions Factors; no source is zero over its life cycle, some are low-carbon

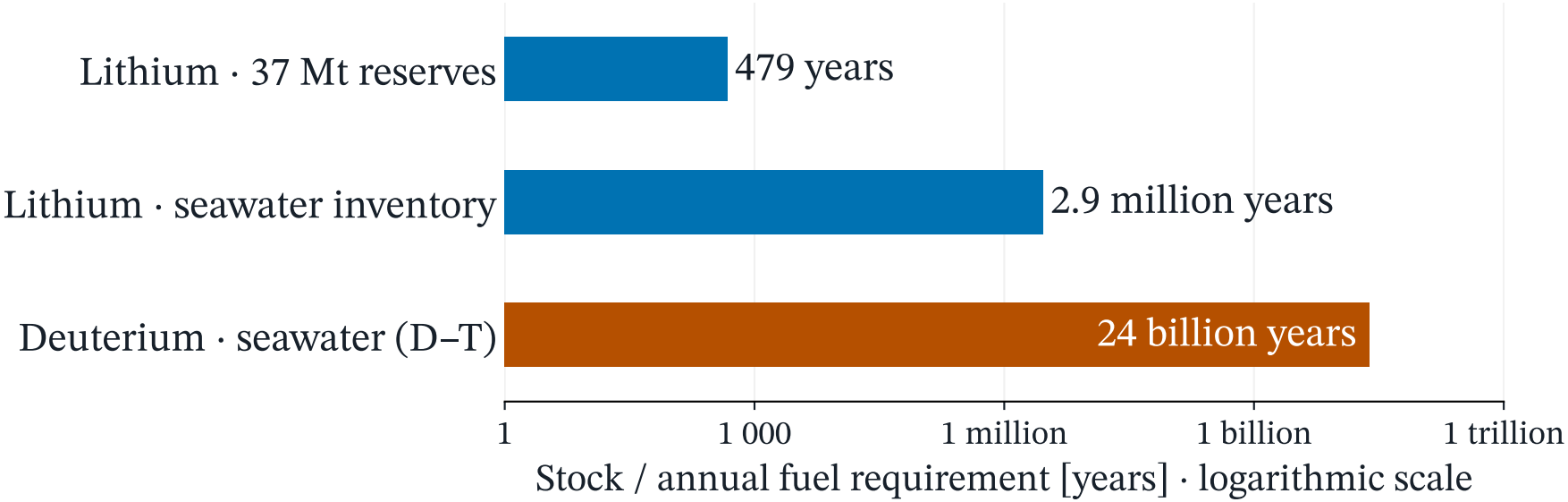


EUROfusion, Wikimedia Commons, CC BY 4.0

0.11 Fusion fuel reserves

World 2024 primary energy matched as electricity: 639 EJ/year

$$T = \frac{\text{stock} \times \text{electricity per kg}}{\text{electricity per year}}$$



MacKay · extraction / fusion: 2.5 / 2300 ≈ 0.1 %

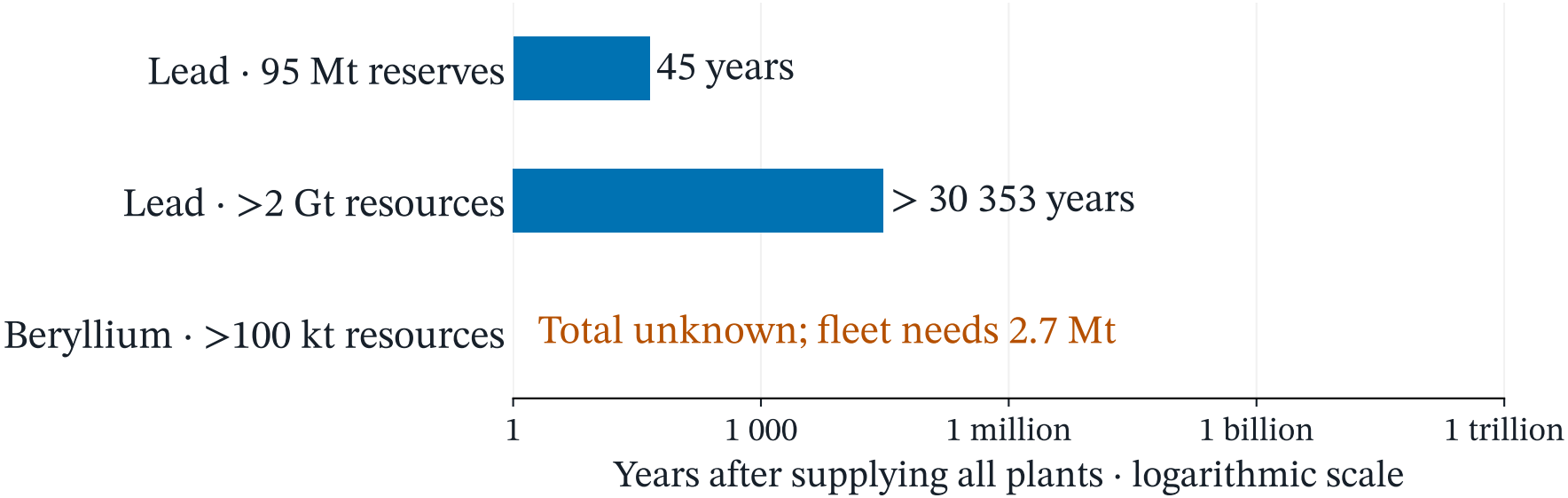
Laboratory extraction demonstrated · industrial scale unproven

EIA 2024 · USGS 2026 · NOAA/NIST · MacKay ch. 24 · Li et al. 2021 (doi:10.1039/D1EE00354B)
Li yield: 2300 kWh(e)/g, assumed · D–T: 40% conversion, still needs Li · no growth, fuel losses or competing uses

0.11 Lead and beryllium

Fusion electricity: 639 EJ/year

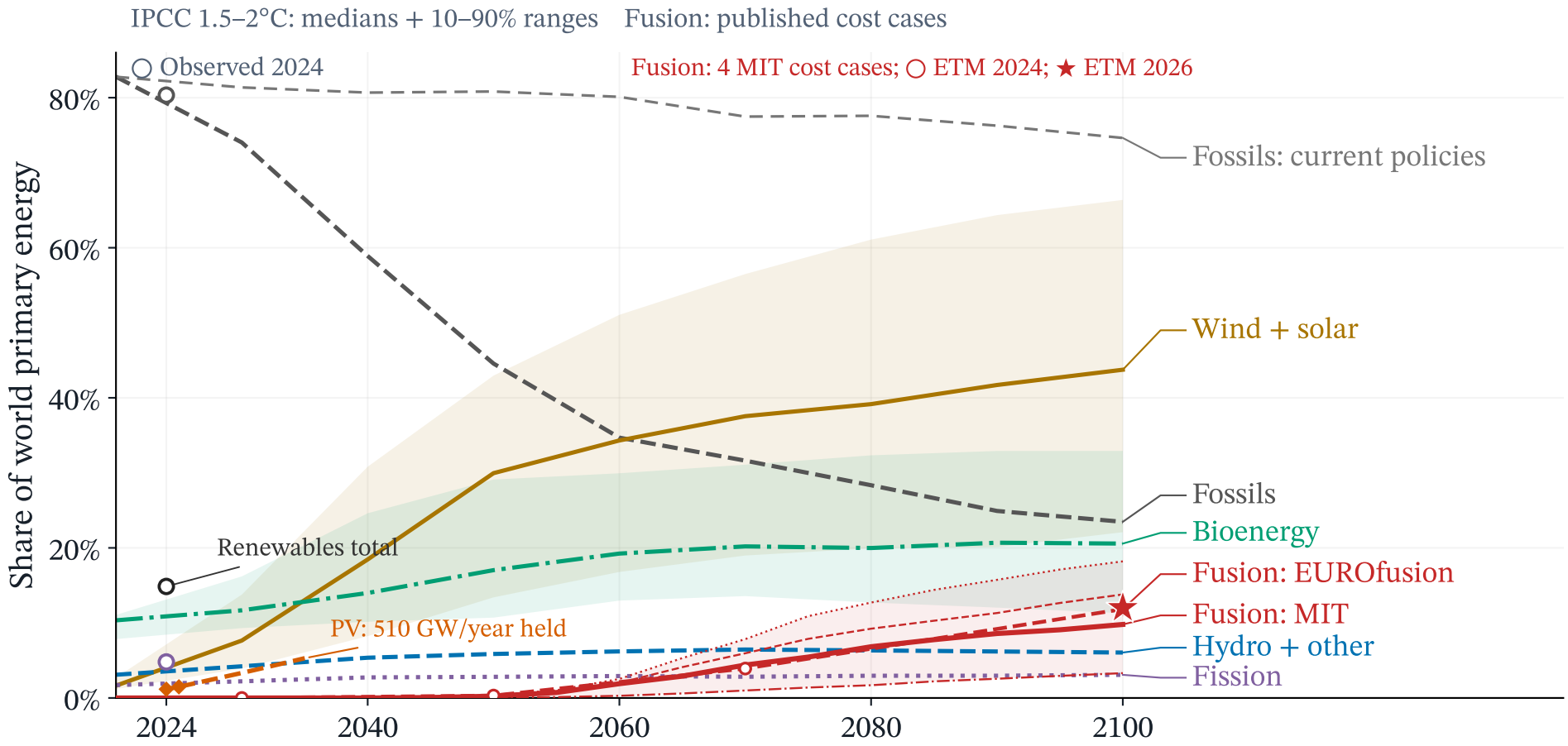
$$T = \frac{\text{stock} - \text{material in plants}}{\text{annual consumption}}$$



Initially in all plants: 92 Mt lead or 2.7 Mt beryllium

Bradshaw et al. 2011 · USGS 2026 · 639 EJ/year: world primary energy, 2024
90% capacity factor · full recycling · constant demand · no other uses

0.12 Energy to 2100



IPCC AR6: 356 pathways · current-policy assumptions: 2021 · IEA observed 2024
Fusion: published output ÷ AR6 median primary demand · PV guide: 2025 build rate, 15% capacity factor

Credits

2 NASA Earth Observatory, Black Marble 2016, public domain
science.nasa.gov/earth/earth-observatory/earth-at-night

4 Dollar Street, Gapminder, CC BY 4.0
gapminder.org/dollar-street

11 Naturpuur, Windpark Freiländeralm, CC BY-SA 4.0
commons.wikimedia.org

21 EUROfusion, JET vessel internal view, CC BY 4.0
commons.wikimedia.org/wiki/File:JET_vessel_internal_view.jpg

Data: EIA and USGS (public domain); World Bank WDI (CC BY 4.0); Statistics Austria; BGR; IAEA/NEA (CC BY 4.0); PVGIS (EU JRC); Our World in Data (CC BY 4.0); FAO; USDA; DOE/NREL/ALLIANCE; Loss et al.; Copernicus Sentinel-2; Stadt Graz OGD (CC BY 4.0).
Plots and calculations: Christopher Albert, CC BY 4.0. Method after D. MacKay, Sustainable Energy — without the hot air (2009).

AI tools assisted drafting and layout; scientific review by Christopher Albert.